

Frogger's Random Hop

The Situation: Frogger wakes up from a nap and finds himself on a log floating out to sea. Taking notice of his surroundings, he observes that behind him is one log and an endless sea. In front of him are two logs and the safety of the shoreline. At some point during his nap, his legs were splashed by salt water, making his hops unpredictable. Every second, Frogger hops either forward or backward with an equal probability of $\frac{1}{2}$. What is the probability that he hops to shore?

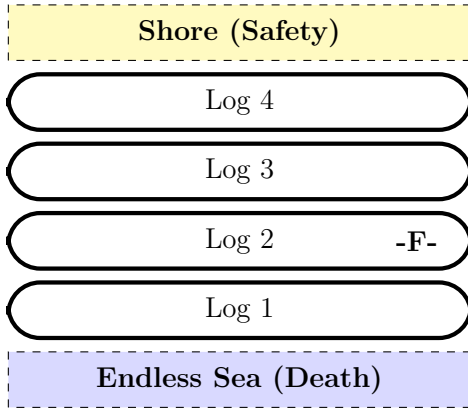


Figure 1: Visual mapping of the random hop environment. The **-F-** marker denotes Frogger's starting position as he wakes up and attempts to reach the Shoreline.

Solution

We begin our solution to this problem by first noticing that Frogger makes it to the shore only when the following criteria occur:

- Frogger makes it from Log 2 to Log 3 without jumping into the Sea.
- Frogger makes it from Log 3 to Log 4 without jumping into the Sea.
- Frogger makes it from Log 4 to Shore without jumping into the Sea.

Let the probabilities of these events occurring be denoted ϕ_2, ϕ_3, ϕ_4 respectively. Since these events are independent, the probability that Frogger survives is given by:

$$\begin{aligned} P(\text{Frogger Survives}) &= P(\text{Log 2 to Log 3}) \cdot P(\text{Log 3 to Log 4}) \cdot P(\text{Log 4 to Shore}) \\ &= \phi_2 \cdot \phi_3 \cdot \phi_4 \end{aligned}$$

Using an Infinite Geometric Series to find the Solution:

Consider what happens in the general case. Frogger is currently on log k . We want to find the probability, ϕ_k , that he makes it to log $k + 1$ without jumping into the sea. Consider the two possible series of events that must occur when Frogger hops:

- Frogger hops forward to log $k + 1$ with probability $\frac{1}{2}$.
- Frogger hops backward to log $k - 1$ with probability $\frac{1}{2}$. In this case, frogger must now get back to log k (with probability ϕ_{k-1}), and then try again to reach log $k + 1$ (with probability ϕ_k).

That is, to fulfill the criteria for making it from log k to log $k + 1$, without jumping into the sea, two events may occur. Either Frogger hops forward immediately, or he hops backward, recovers to log k and then successfully hops forward. Crucially, for any given log k , we have a recursive identity between ϕ_k and ϕ_{k-1} given by:

$$\begin{aligned}\phi_k &= \underbrace{P(\text{Forward Hop})}_{\text{Immediately hops forward}} + \underbrace{P(\text{Backward Hop}) \cdot P(\text{Forward Recovery Hop}) \cdot P(\text{Forward Hop})}_{\text{Hops back, but still makes it forward}} \\ &= \underbrace{\frac{1}{2}}_{\text{Immediate Success}} + \underbrace{\frac{1}{2} \cdot \phi_{k-1} \cdot \phi_k}_{\text{Success via Recovery}}\end{aligned}$$

Solving for ϕ_k in terms of ϕ_{k-1} , we have:

$$\begin{aligned}\phi_k - \frac{1}{2}\phi_{k-1}\phi_k &= \frac{1}{2} \\ \phi_k \left(1 - \frac{1}{2}\phi_{k-1}\right) &= \frac{1}{2} \\ \phi_k &= \frac{\frac{1}{2}}{1 - \frac{1}{2}\phi_{k-1}}\end{aligned}$$

Note that we have uncovered an infinite geometric series with first term $\frac{1}{2}$ and common ratio $\frac{1}{2}\phi_{k-1}$. In summation form, we can find the exact probability ϕ_k as follows:

$$\begin{aligned}\phi_k &= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2}\phi_{k-1} + \left(\frac{1}{2} \cdot \frac{1}{2}\phi_{k-1}\right)^2 + \left(\frac{1}{2} \cdot \frac{1}{2}\phi_{k-1}\right)^3 + \dots \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2} \cdot \frac{1}{2}\phi_{k-1}\right)^n \cdot \frac{1}{2} \\ &= \frac{1}{2} \sum_{n=0}^{\infty} \left(\frac{1}{2}\phi_{k-1}\right)^n\end{aligned}$$

Finding the Probability that Frogger Survives:

To find the probability that Frogger survives, we need to find ϕ_2, ϕ_3, ϕ_4 . We can use the recursive relationship we found above to find these values. Note that $\phi_0 = 0$, since if you are in the sea, the probability of reaching the shore is zero by the definition of the problem. We use the sum of an infinite geometric series to find the following probabilities:

$$\phi_1 = \frac{\frac{1}{2}}{1 - \frac{1}{2}(0)} = \frac{1}{2}$$

$$\phi_2 = \frac{\frac{1}{2}}{1 - \frac{1}{2}\left(\frac{1}{2}\right)} = \frac{1/2}{3/4} = \frac{2}{3}$$

$$\phi_3 = \frac{\frac{1}{2}}{1 - \frac{1}{2}\left(\frac{2}{3}\right)} = \frac{1/2}{2/3} = \frac{3}{4}$$

$$\phi_4 = \frac{\frac{1}{2}}{1 - \frac{1}{2}\left(\frac{3}{4}\right)} = \frac{1/2}{5/8} = \frac{4}{5}$$

Since $P(\text{Frogger Survives}) = \phi_2 \cdot \phi_3 \cdot \phi_4$, we have:

$$P(\text{Frogger Survives}) = \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} = \frac{2}{5}$$

Thus, Frogger makes it to shore with a probability of 40%.